

Numbers, Percents & Ratios

ANSWER KEY



Properties of integers and number classification

- 1 Classify the number $-\frac{7}{2}$ as an integer, rational number, or irrational number.
 $-\frac{7}{2} = -3.5$ is a rational number (it can be expressed as a ratio of two integers), but it is NOT an integer, since it is not a whole number.
- 2 Find the greatest common factor (GCF) and least common multiple (LCM) of 18 and 24.
 $18 = 2 \times 3^2$, $24 = 2^3 \times 3$. GCF = $2 \times 3 = 6$. LCM = $2^3 \times 3^2 = 8 \times 9 = 72$.

Divisibility and prime factorization

- 3 Find the prime factorization of 360.
 $360 = 2^3 \times 3^2 \times 5$.
- 4 Determine whether 91 is prime, and justify your answer.
 $91 = 7 \times 13$, so 91 is not prime — it has factors other than 1 and itself.

Percent increase and decrease

- 5 A price increases from 80 dollars to 92 dollars. Find the percent increase.
Increase: $92 - 80 = 12$. Percent increase: $\frac{12}{80} \times 100\% = 15\%$.
- 6 A population of 5000 decreases by 8%. Find the new population.
Decrease: $5000 \times 0.08 = 400$. New population: $5000 - 400 = 4600$.

Percent of a percent (successive percent changes)

- 7 A jacket is marked down 30%, then an additional 10% is taken off the sale price. If the original price was 100 dollars, find the final price.
After the first discount: $100(0.70) = 70$ dollars. After the second discount: $70(0.90) = 63$ dollars.

Ratios and proportional reasoning

- 8 The ratio of boys to girls in a class is 3:5. If there are 40 students total, find the number of boys.

Total ratio parts: $3 + 5 = 8$. Each part represents $\frac{40}{8} = 5$ students. Boys: $3 \times 5 = 15$.

- 9 A recipe uses a ratio of 2 cups flour to 3 cups sugar. If a baker uses 15 cups of sugar, how much flour is needed?

Set up a proportion: $\frac{2}{3} = \frac{x}{15}$. Cross-multiply: $3x = 30$, so $x = 10$ cups of flour.

Scientific notation

- 10 Write 0.000452 in scientific notation.

$$4.52 \times 10^{-4}.$$

- 11 Evaluate $(3 \times 10^4) \times (2 \times 10^{-2})$, giving your answer in scientific notation.

$$(3 \times 2) \times 10^{4+(-2)} = 6 \times 10^2.$$

Absolute value and number line reasoning

- 12 Find all values of x satisfying $|x - 4| = 9$.

Either $x - 4 = 9$, giving $x = 13$, or $x - 4 = -9$, giving $x = -5$.

- 13 Find the distance between -8 and 15 on the number line, using absolute value.

$$\text{Distance} = |15 - (-8)| = |23| = 23.$$

Word problems involving rates and mixtures

- 14 A store mixes 4 pounds of candy costing 3 dollars per pound with 6 pounds of candy costing 5 dollars per pound. Find the average price per pound of the mixture.

Total cost: $4(3) + 6(5) = 12 + 30 = 42$ dollars. Total weight: $4 + 6 = 10$ pounds. Average price:
 $\frac{42}{10} = 4.20$ dollars per pound.

- 15 A factory's production increased from 240 units per day to 300 units per day. Express this change as a ratio in simplest form.

$$\text{Ratio of new to old: } \frac{300}{240} = \frac{5}{4}.$$

Combining percent, ratio, and rate concepts

- 16** This question has two parts. A store buys a product for 40 dollars and marks it up 60% to set the retail price. (a) Find the retail price. (b) If the store later offers a 25% discount off the retail price, find the discounted price, and determine whether the store still makes a profit compared to its 40-dollar cost.

(a) Markup amount: $40(0.60) = 24$ dollars. Retail price: $40 + 24 = 64$ dollars. (b) Discount amount: $64(0.25) = 16$ dollars. Discounted price: $64 - 16 = 48$ dollars. Since $48 > 40$, the store still makes a profit of $48 - 40 = 8$ dollars per item.

Order of operations and simplifying numerical expressions

- 17** Evaluate $6 + 2(9 - 3)^2 \div 4$.

Parentheses first: $9 - 3 = 6$. Exponent: $6^2 = 36$. Then: $6 + 2(36) \div 4 = 6 + 72 \div 4 = 6 + 18 = 24$.

- 18** Evaluate $\frac{5^2 - 3^2}{2^3 - 2}$.

Numerator: $25 - 9 = 16$. Denominator: $8 - 2 = 6$. Result: $\frac{16}{6} = \frac{8}{3}$.

Fractions, decimals, and conversions

- 19** Convert $\frac{7}{8}$ to a decimal and to a percent.

$\frac{7}{8} = 0.875 = 87.5\%$.

- 20** Add $\frac{2}{3} + \frac{3}{4}$, giving your answer as a simplified fraction.

Common denominator 12: $\frac{8}{12} + \frac{9}{12} = \frac{17}{12}$.

Sequences: arithmetic and geometric

- 21** Find the 10th term of the arithmetic sequence 4, 9, 14, 19, ...

First term $a_1 = 4$, common difference $d = 5$. $a_{10} = a_1 + 9d = 4 + 45 = 49$.

- 22** Find the 6th term of the geometric sequence 3, 6, 12, 24, ...

First term $a_1 = 3$, common ratio $r = 2$. $a_6 = a_1 \cdot r^5 = 3(32) = 96$.

A multi-step number theory and ratio problem

- 23** This question has two parts. A recipe for 12 cookies requires a ratio of 3 cups flour to 2 cups sugar to 1 cup butter. (a) Find the amount of each ingredient needed to make 30 cookies. (b) If flour costs 0.60 dollars per cup, sugar costs 0.80 dollars per cup, and butter costs 1.50 dollars per cup, find the total ingredient cost for 30 cookies.

(a) Scale factor: $\frac{30}{12} = 2.5$. Flour: $3 \times 2.5 = 7.5$ cups. Sugar: $2 \times 2.5 = 5$ cups. Butter: $1 \times 2.5 = 2.5$ cups. (b) Total cost: $7.5(0.60) + 5(0.80) + 2.5(1.50) = 4.5 + 4 + 3.75 = 12.25$ dollars.

Multiple choice — easy

- 24** The price of a backpack decreases from 260 dollars to 26 dollars. Find the percent decrease.

- a. 85%
- b. 90% ✓
- c. 95%
- d. 100%

Decrease: $260 - 26 = 234$ dollars. Percent decrease: $234/260 \times 100\% = 90\%$.

- 25** The price of a jacket decreases from 280 dollars to 182 dollars. Find the percent decrease.

- a. 30%
- b. 45%
- c. 35% ✓
- d. 40%

Decrease: $280 - 182 = 98$ dollars. Percent decrease: $98/280 \times 100\% = 35\%$.

- 26** The price of a television decreases from 160 dollars to 128 dollars. Find the percent decrease.

- a. 30%
- b. 25%
- c. 20% ✓
- d. 15%

Decrease: $160 - 128 = 32$ dollars. Percent decrease: $32/160 \times 100\% = 20\%$.

27 The price of a phone increases from 240 dollars to 372 dollars. Find the percent increase.

- a. 60%
- b. 65%
- c. 50%
- d. 55% ✓

Increase: $372 - 240 = 132$ dollars. Percent increase: $132/240 \times 100\% = 55\%$.

28 The price of a desk decreases from 180 dollars to 144 dollars. Find the percent decrease.

- a. 15%
- b. 20% ✓
- c. 25%
- d. 30%

Decrease: $180 - 144 = 36$ dollars. Percent decrease: $36/180 \times 100\% = 20\%$.

29 The price of a television decreases from 200 dollars to 180 dollars. Find the percent decrease.

- a. 10% ✓
- b. 20%
- c. 15%
- d. 5%

Decrease: $200 - 180 = 20$ dollars. Percent decrease: $20/200 \times 100\% = 10\%$.

30 The price of a phone decreases from 220 dollars to 209 dollars. Find the percent decrease.

- a. 20%
- b. 15%
- c. 5% ✓
- d. 10%

Decrease: $220 - 209 = 11$ dollars. Percent decrease: $11/220 \times 100\% = 5\%$.

31 The price of a jacket decreases from 100 dollars to 95 dollars. Find the percent decrease.

- a. 10%
- b. 5% ✓
- c. 20%
- d. 15%

Decrease: $100 - 95 = 5$ dollars. Percent decrease: $5/100 \times 100\% = 5\%$.

32 The price of a desk decreases from 280 dollars to 266 dollars. Find the percent decrease.

- a. 10%
- b. 15%
- c. 5% ✓
- d. 20%

Decrease: $280 - 266 = 14$ dollars. Percent decrease: $14/280 \times 100\% = 5\%$.

33 The price of a jacket decreases from 240 dollars to 156 dollars. Find the percent decrease.

- a. 45%
- b. 30%
- c. 40%
- d. 35% ✓

Decrease: $240 - 156 = 84$ dollars. Percent decrease: $84/240 \times 100\% = 35\%$.

34 The price of a laptop decreases from 60 dollars to 30 dollars. Find the percent decrease.

a. 50% ✓

b. 60%

c. 45%

d. 55%

Decrease: $60 - 30 = 30$ dollars. Percent decrease: $30/60 \times 100\% = 50\%$.

35 The price of a desk increases from 240 dollars to 396 dollars. Find the percent increase.

a. 75%

b. 65% ✓

c. 70%

d. 60%

Increase: $396 - 240 = 156$ dollars. Percent increase: $156/240 \times 100\% = 65\%$.

36 The price of a television decreases from 260 dollars to 104 dollars. Find the percent decrease.

a. 55%

b. 70%

c. 60% ✓

d. 65%

Decrease: $260 - 104 = 156$ dollars. Percent decrease: $156/260 \times 100\% = 60\%$.

37 The price of a jacket decreases from 160 dollars to 48 dollars. Find the percent decrease.

- a. 65%
- b. 80%
- c. 70% ✓
- d. 75%

Decrease: $160 - 48 = 112$ dollars. Percent decrease: $112/160 \times 100\% = 70\%$.

38 The price of a laptop increases from 300 dollars to 375 dollars. Find the percent increase.

- a. 35%
- b. 20%
- c. 30%
- d. 25% ✓

Increase: $375 - 300 = 75$ dollars. Percent increase: $75/300 \times 100\% = 25\%$.

Multiple choice — medium

39 The ratio of novels to textbooks in a group is 2:1. If there are 33 books total, find the number of novels.

- a. 22 ✓
- b. 33
- c. 24
- d. 11

Total ratio parts: $2 + 1 = 3$. Each part represents $33/3 = 11$ books. Novels: $2 \times 11 = 22$.

40 The ratio of novels to textbooks in a group is 3:5. If there are 48 books total, find the number of novels.

- a. 30
- b. 18 ✓
- c. 24
- d. 12

Total ratio parts: $3 + 5 = 8$. Each part represents $48/8 = 6$ books. Novels: $3 \times 6 = 18$.

41 The ratio of cats to dogs in a group is 3:2. If there are 50 pets total, find the number of cats.

- a. 32
- b. 40
- c. 30 ✓
- d. 20

Total ratio parts: $3 + 2 = 5$. Each part represents $50/5 = 10$ pets. Cats: $3 \times 10 = 30$.

42 The ratio of red marbles to blue marbles in a group is 4:7. If there are 55 marbles total, find the number of red marbles.

- a. 25
- b. 20 ✓
- c. 15
- d. 35

Total ratio parts: $4 + 7 = 11$. Each part represents $55/11 = 5$ marbles. Red marbles: $4 \times 5 = 20$.

43 The ratio of novels to textbooks in a group is 3:5. If there are 40 books total, find the number of novels.

- a. 20
- b. 10
- c. 15 ✓
- d. 25

Total ratio parts: $3 + 5 = 8$. Each part represents $40/8 = 5$ books. Novels: $3 \times 5 = 15$.

44 The ratio of sedans to trucks in a group is 4:3. If there are 21 vehicles total, find the number of sedans.

- a. 15
- b. 14
- c. 12 ✓
- d. 9

Total ratio parts: $4 + 3 = 7$. Each part represents $21/7 = 3$ vehicles. Sedans: $4 \times 3 = 12$.

45 The ratio of red marbles to blue marbles in a group is 5:4. If there are 54 marbles total, find the number of red marbles.

- a. 30 ✓
- b. 36
- c. 26
- d. 24

Total ratio parts: $5 + 4 = 9$. Each part represents $54/9 = 6$ marbles. Red marbles: $5 \times 6 = 30$.

46 The ratio of boys to girls in a group is 5:9. If there are 28 students total, find the number of boys.

a. 10 ✓

b. 12

c. 8

d. 18

Total ratio parts: $5 + 9 = 14$. Each part represents $28/14 = 2$ students. Boys: $5 \times 2 = 10$.

47 The ratio of cats to dogs in a group is 2:3. If there are 60 pets total, find the number of cats.

a. 36

b. 12

c. 21

d. 24 ✓

Total ratio parts: $2 + 3 = 5$. Each part represents $60/5 = 12$ pets. Cats: $2 \times 12 = 24$.

48 The ratio of sedans to trucks in a group is 1:3. If there are 16 vehicles total, find the number of sedans.

a. 4 ✓

b. 12

c. 1

d. 8

Total ratio parts: $1 + 3 = 4$. Each part represents $16/4 = 4$ vehicles. Sedans: $1 \times 4 = 4$.

- 49 The ratio of red marbles to blue marbles in a group is 7:5. If there are 108 marbles total, find the number of red marbles.
- a. 54
 - b. 45
 - c. 63 ✓
 - d. 72

Total ratio parts: $7 + 5 = 12$. Each part represents $108/12 = 9$ marbles. Red marbles: $7 \times 9 = 63$.

- 50 The ratio of boys to girls in a group is 3:1. If there are 40 students total, find the number of boys.
- a. 10
 - b. 40
 - c. 30 ✓
 - d. 20

Total ratio parts: $3 + 1 = 4$. Each part represents $40/4 = 10$ students. Boys: $3 \times 10 = 30$.

- 51 The ratio of cats to dogs in a group is 7:5. If there are 60 pets total, find the number of cats.
- a. 25
 - b. 35 ✓
 - c. 30
 - d. 40

Total ratio parts: $7 + 5 = 12$. Each part represents $60/12 = 5$ pets. Cats: $7 \times 5 = 35$.

52 The ratio of boys to girls in a group is 9:7. If there are 80 students total, find the number of boys.

- a. 40
- b. 50
- c. 45 ✓
- d. 35

Total ratio parts: $9 + 7 = 16$. Each part represents $80/16 = 5$ students. Boys: $9 \times 5 = 45$.

53 The ratio of sedans to trucks in a group is 6:5. If there are 88 vehicles total, find the number of sedans.

- a. 48 ✓
- b. 40
- c. 56
- d. 43

Total ratio parts: $6 + 5 = 11$. Each part represents $88/11 = 8$ vehicles. Sedans: $6 \times 8 = 48$.

Multiple choice — hard

54 Find the greatest common factor (GCF) of 189 and 21.

- a. 10
- b. 21 ✓
- c. 189
- d. 42

$189 = 3^3 \times 7$, $21 = 3 \times 7$. GCF = $3 \times 7 = 21$.

55 Find the greatest common factor (GCF) of 18 and 12.

a. 6 ✓

b. 3

c. 36

d. 12

$$18 = 2 \times 3^2, 12 = 2^2 \times 3. \text{ GCF} = 2 \times 3 = 6.$$

56 Find the greatest common factor (GCF) of 500 and 200.

a. 100 ✓

b. 200

c. 50

d. 1000

$$500 = 2^2 \times 5^3, 200 = 2^3 \times 5^2. \text{ GCF} = 2^2 \times 5^2 = 100.$$

57 Find the greatest common factor (GCF) of 24 and 54.

a. 3

b. 6 ✓

c. 216

d. 12

$$24 = 3 \times 2^3, 54 = 3^3 \times 2. \text{ GCF} = 3 \times 2 = 6.$$

58 Find the greatest common factor (GCF) of 147 and 21.

a. 42

b. 147

c. 10

d. 21 ✓

$$147 = 3 \times 7^2, 21 = 3 \times 7. \text{ GCF} = 3 \times 7 = 21.$$

59 Find the 10th term of the arithmetic sequence 2, 1, 0, -1, ...

- a. -8
- b. -7 ✓
- c. -6
- d. -10

First term $a_1 = 2$, common difference $d = -1$. $a_{10} = a_1 + 9d = 2 + (9)(-1) = -7$.

60 Find the 16th term of the arithmetic sequence -19, -24, -29, -34, ...

- a. -94 ✓
- b. -89
- c. -91
- d. -99

First term $a_1 = -19$, common difference $d = -5$. $a_{16} = a_1 + 15d = -19 + (15)(-5) = -94$.

61 Find the 20th term of the arithmetic sequence 19, 23, 27, 31, ...

- a. 87
- b. 95 ✓
- c. 99
- d. 91

First term $a_1 = 19$, common difference $d = 4$. $a_{20} = a_1 + 19d = 19 + (19)(4) = 95$.

62 Find the 9th term of the arithmetic sequence $-17, -21, -25, -29, \dots$

- a. -49 ✓
- b. -52
- c. -45
- d. -53

First term $a_1 = -17$, common difference $d = -4$. $a_9 = a_1 + 8d = -17 + (8)(-4) = -49$.

63 Find the 8th term of the arithmetic sequence $-3, -5, -7, -9, \dots$

- a. -21
- b. -15
- c. -19
- d. -17 ✓

First term $a_1 = -3$, common difference $d = -2$. $a_8 = a_1 + 7d = -3 + (7)(-2) = -17$.

64 Evaluate $(3 \times 10^{-1}) \times (9 \times 10^{-2})$, giving your answer in scientific notation.

- a. 2.7×10^{-3}
- b. 2.7×10^{-1}
- c. 27×10^{-3}
- d. 2.7×10^{-2} ✓

$$(3 \times 9) \times 10^{-1-2} = 27 \times 10^{-3} = 2.7 \times 10^{-2}.$$

65 Evaluate $(8 \times 10^6) \times (4 \times 10^2)$, giving your answer in scientific notation.

- a. 32×10^8
- b. 3.2×10^8
- c. 3.2×10^9 ✓
- d. 3.2×10^{10}

$$(8 \times 4) \times 10^{6+2} = 32 \times 10^8 = 3.2 \times 10^9.$$

66 Evaluate $(7 \times 10^{-1}) \times (2 \times 10^1)$, giving your answer in scientific notation.

- a. 14×10^0
- b. 1.4×10^2
- c. 1.4×10^0
- d. 1.4×10^1 ✓

$$(7 \times 2) \times 10^{-1+1} = 14 \times 10^0 = 1.4 \times 10^1.$$

67 Evaluate $(6 \times 10^{-6}) \times (2 \times 10^1)$, giving your answer in scientific notation.

- a. 1.2×10^{-5}
- b. 1.2×10^{-3}
- c. 12×10^{-5}
- d. 1.2×10^{-4} ✓

$$(6 \times 2) \times 10^{-6+1} = 12 \times 10^{-5} = 1.2 \times 10^{-4}.$$

68 Evaluate $(3 \times 10^{-6}) \times (3 \times 10^5)$, giving your answer in scientific notation.

- a. 9×10^0
- b. 9×10^{-1} ✓
- c. 9×10^{-2}
- d. 9×10^{-3}

$$(3 \times 3) \times 10^{-6+5} = 9 \times 10^{-1} = 9 \times 10^{-1}.$$